

Ex.no:6b	DATA VISUALIZATION USING POWER BI (ADVANCED SALES ANALYTICS)

Aim:

To load the **Life Expectancy dataset** into Power BI, transform and clean the data, and visualize it to obtain meaningful insights about **life expectancy, country status, GDP, mortality, and schooling**.

Algorithm:

ALGORITHM 1: IMPORTING DATA INTO POWER BI

Step 1

Open Power BI Desktop

Step 2

Select:

Home → Get Data

Step 3

Choose Excel or CSV

Step 4

Browse dataset

Step 5

Click Load

Step 6

Verify imported data

ALGORITHM 2: DATA TRANSFORMATION USING POWER QUERY

Step 1

Select **Transform Data**

Step 2

Open Power Query Editor

Step 3

Remove duplicates

Step 4

Handle null values in numerical columns such as **Life expectancy, Adult Mortality, GDP, Population, and Schooling**.

Step 5

Rename columns & format text casing (e.g., **Year, Life expectancy, GDP, Population, and Schooling**.)

Step 6

Apply changes & Close

ALGORITHM 3: ADVANCED TRANSFORMATIONS & DATE PARSING

Step 1

Select **Country** column and ensure that its data type is set to **Text**.

Step 2

Select the **Status** column and ensure that its data type is set to **Text**.

Step 3

Select numerical columns such as **Life expectancy, Adult Mortality, GDP, Population, and Schooling**.

Step 4

Set their data type to **Decimal Number** or **Whole Number** as appropriate.

Step 5

Replace or remove null values from numerical columns.

Step 6

Click **Close & Apply** to load the clean dataset into the Power BI model.

ALGORITHM 4: CREATING CALCULATED COLUMN (REVENUE)

Step 1

Navigate to **Report View** or **Table View**

Step 2

Right- click the Life Expectancy dataset table and select **New Column**.

Step 3

Enter the calculated column formula:

Life Expectancy Category =

IF(

'Life Expectancy Data'[Life expectancy] >= 70,

"High",

"Low"

)

Step 4

Press **Enter** to validate and calculate the values.

ALGORITHM 5: DASHBOARD VISUALIZATION & REPORTING

Step 1

Select **Line Chart** visualization and map **Year** to the X-axis and **Average of Life expectancy** to the Y-axis.

Step 2

Select a **Bar / Column Chart** and map **Status** to the X-axis and **Average Life Expectancy** to the Y-axis.

Step 3

Select a **Donut Chart** to display the distribution of countries based on **Status**.

Step 4

Add **Slicer** visuals for **Country, Year, and Status** to enable interactive filtering.

Step 5

Add suitable **Card visuals** to display values such as **Average Life Expectancy, Number of Countries, and Average GDP**.

Step 6

Add suitable titles and labels to the visualizations.

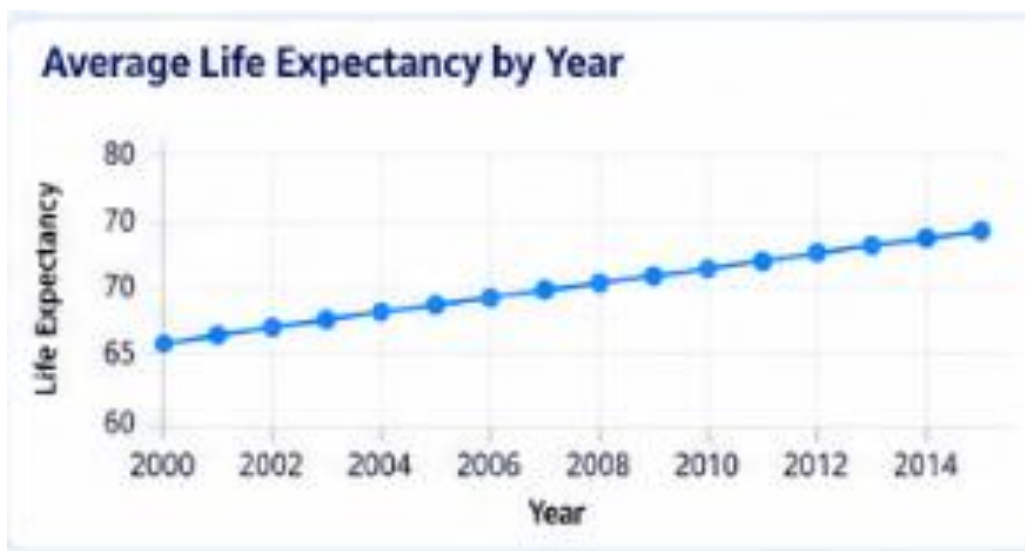
Step 7

Save the Power BI file (**.pbix**).

Output:



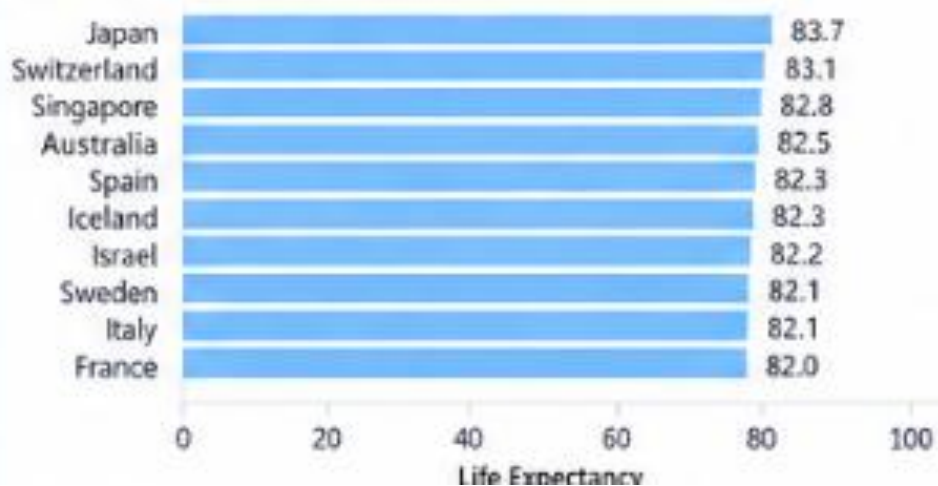
Country	Year	Status	Life expectancy	Adult Mortality	GDP	Schooling
Afghanistan	2015	Developing	65.0	263.0	584.259	10.1
Afghanistan	2014	Developing	59.9	271.0	612.696	10.0
Afghanistan	2013	Developing	59.9	268.0	631.745	9.9
Afghanistan	2012	Developing	59.5	272.0	669.959	9.8
Afghanistan	2011	Developing	59.2	275.0	63.537	9.5
Albania	2015	Developing	77.8	74.0	3954.227	12.0
Albania	2014	Developing	77.5	77.0	4579.562	11.9
Albania	2013	Developing	77.2	80.0	4414.723	11.8
Albania	2012	Developing	76.9	83.0	4241.175	11.6
Albania	2011	Developing	76.7	84.0	4127.117	11.4

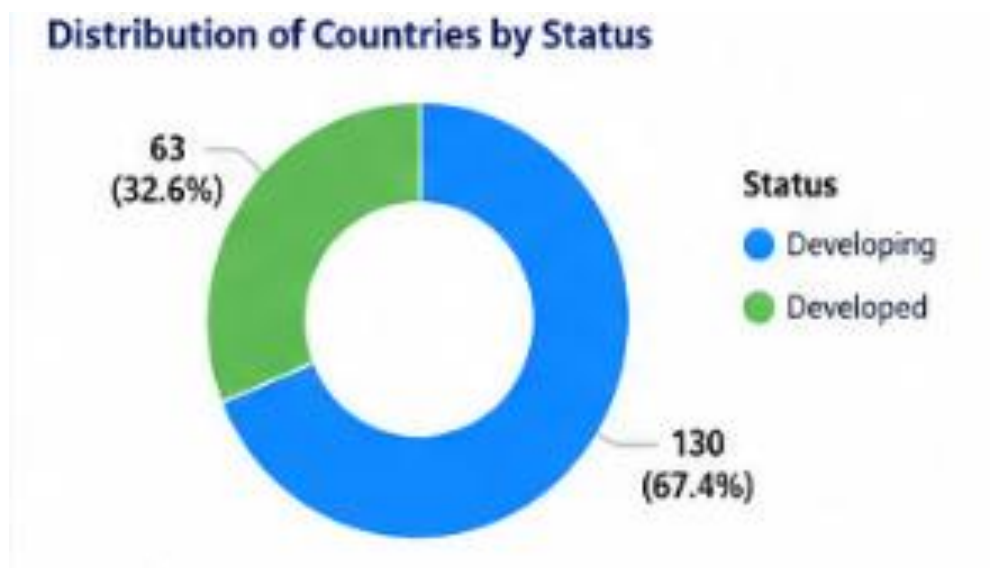


Average Life Expectancy by Country Status



Top 10 Countries by Life Expectancy (Latest Year)





Result: